

MIAE 221: Materials Science for Engineers

Master Course Overview, Study Roadmap & Exam Strategy

1. Course Information & Key Contacts

- **Course Code:** MIAE 221 (formerly MECH 221) — *Materials Science for Engineers*
- **Department:** Department of Mechanical, Industrial & Aerospace Engineering (MIAE)
- **Institution:** Concordia University, Gina Cody School of Engineering and Computer Science
- **Instructor: Dr. Mamoun Medraj, P.Eng**
 - **Email:** mamoun.medraj@concordia.ca
 - **Office:** Room EV 12.185
- **Required Textbook:**
 - *Materials Science and Engineering: An Introduction*, William D. Callister, Jr. & David G. Rethwisch (Wiley, 10th / 9th Edition).
- **Teaching Assistants (TAs):**
 - Lama Mahmoud (lama.mahmoud@mail.concordia.ca)
 - Michael Pudlo (michael.pudlo@mail.concordia.ca)
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2. Course Assessment & Grading Scheme

The course evaluates conceptual mastery, quantitative derivation skills, and microstructural understanding:

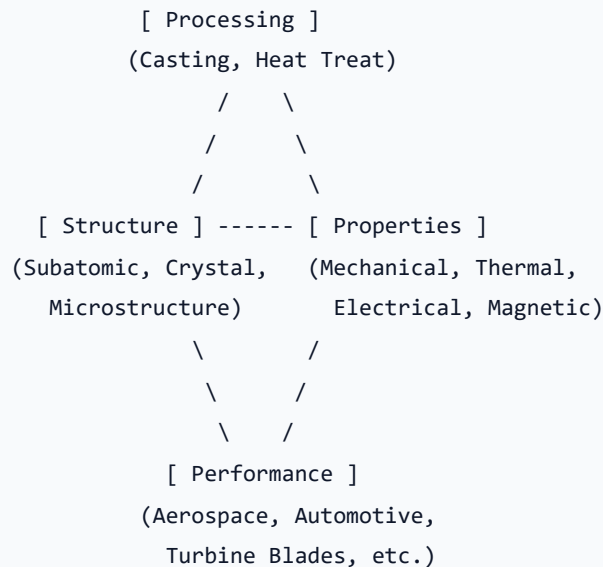
Assessment Component	Weight	Key Logistics & Strategic Tips
Tutorial Quizzes & In-Class Tests	~15–20%	Administered during weekly tutorials. Problems are directly derived from the assigned Practice Problem Sets. Master the problem sets to secure full points here.
Laboratory Component & Reports	~15%	Hands-on metallurgical mounting, polishing, tensile testing, and hardness measurements (Rockwell / Brinell). Rigorous error analysis is expected.
Midterm Examination	~25–30%	Scheduled for Friday, October 30th. Covers Chapters 1 through 7 (Atomic Structure, Bonding, Crystal Structures, Imperfections, and Diffusion).
Final Examination	~40–50%	Comprehensive 3-hour exam covering the entire semester (including Phase Diagrams, TTT curves, Heat Treatment, and Mechanical Properties).

[!IMPORTANT] **Assignments Policy:** While problem sets are not formally collected for grading, their exact concepts, equations, and numerical structures appear in tutorial quizzes and midterm exams. Solving every

problem independently is the single most effective predictor of success in MIAE 221.

3. The Central Paradigm of Materials Science

Everything in MIAE 221 revolves around the **Materials Tetrahedron**:



The Chain of Logic:

1. **Processing:** How you make it (e.g., rapid cooling vs. slow furnace cooling) changes the...
2. **Structure:** How atoms and crystals are arranged, which dictates the...
3. **Properties:** How the material responds to external mechanical loads, temperature, or electrical fields, which determines its...
4. **Performance:** How long it safely survives in an engineering application (aircraft engine, bridge, biocompatible implant).

4. Semester Topical Roadmap

Phase I: Atomic Architecture & Crystal Structure (Weeks 1–4)

- **Lectures 1 & 2: Materials Classification & Atomic Foundations**
 - Classes of materials: Metals, Ceramics, Polymers, Composites, Semiconductors.
 - Bohr atom vs. Wave-mechanical quantum model (n, l, m_l, m_s quantum numbers).
 - Electron configurations, valence states, periodic trends, electronegativity.
 - Interatomic attractive/repulsive forces and net potential energy curves ($E_N = E_A + E_R$).
- **Lecture 3: Chemical Bonding & Physical Properties**
 - Primary bonds: Ionic, Covalent, Metallic.
 - Secondary bonds: Fluctuating dipoles (van der Waals), permanent dipoles, Hydrogen bonds.

- Potential energy well physics: $E_0 \iff T_m, r_0 \iff$ interatomic spacing, curvature $\iff$ Young's modulus E , asymmetry $\iff$ thermal expansion α .
- **Weeks 3–4: Crystallography & Solid Geometry**
 - Unit cells: FCC, BCC, HCP. Coordination number, Atomic Packing Factor (APF).
 - Theoretical density derivations ($\rho = \frac{nA}{V_c N_A}$).
 - Crystallographic directions $[uvw]$ and planes (hkl) (Miller Indices). Linear and planar densities.

Phase II: Crystal Imperfections & Atomic Motion (Weeks 5–7)

- **Point Defects:** Vacancies, self-interstitials, impurities (substitutional vs. interstitial solid solutions, Hume-Rothery rules).
- **Linear Defects (Dislocations):** Edge and screw dislocations, Burgers vector ($\vec{b}$), slip planes.
- **Interfacial & Bulk Defects:** Grain boundaries, twin boundaries, phase boundaries.
- **Diffusion Mechanisms:** Fick's First Law (steady-state) and Fick's Second Law (transient diffusion, error function solutions $\text{erf}(z)$).

Phase III: Mechanical Behavior & Strengthening Mechanisms (Weeks 8–10)

- **Stress-Strain Response:** Engineering stress/strain (σ, ϵ) vs. True stress/strain (σ_T, ϵ_T).
- **Elastic vs. Plastic Deformation:** Hooke's Law, yield strength (σ_y , 0.002 offset), Ultimate Tensile Strength (UTS), ductility (%EL, %RA), resilience, toughness.
- **Dislocation Motion & Strengthening:** Grain size reduction (Hall-Petch equation), solid solution hardening, strain hardening (cold work), precipitation hardening.

Phase IV: Phase Equilibria & Microstructural Evolution (Weeks 11–13)

- **Phase Diagrams:** Gibbs phase rule, binary isomorphous systems (lever rule, tie-line analysis).
- **Eutectic & Eutectoid Systems:** The Iron-Carbon ($Fe - Fe_3C$) phase diagram (ferrite, austenite, cementite, pearlite).
- **Phase Transformations & Heat Treatment:** TTT diagrams, CCT diagrams, martensite, bainite, tempered martensite.

5. First-Principles Exam Strategy & Common Traps

1. Quantum Numbers & Configurations

- **Trap:** Forgetting that transition metals fill the $4s$ subshell *before* $3d$, but when ionized, they lose $4s$ electrons *first* (e.g., $Fe : [Ar]4s^23d^6 \implies Fe^{2+} : [Ar]3d^6$).
- **Verification:** Sum up all superscripts to ensure they match atomic number Z .

2. Energy Minimization & Force Derivation

- **Trap:** Mixing up signs. Attractive energy is negative ($E_A = -A/r$), repulsive energy is positive ($E_R = +B/r^n$).

- **Force vs. Energy:** Force is the derivative of energy with respect to separation: $F = -\frac{dE}{dr}$. At equilibrium separation r_0 , the net force is zero:

$$F_N(r_0) = 0 \iff \left. \frac{dE_N}{dr} \right|_{r=r_0} = 0$$

3. Percent Ionic Character

- Use Pauling's equation:

$$\% \text{ Ionic Character} = [1 - \exp(-0.25(X_A - X_B)^2)] \times 100\%$$

- Ensure electronegativities X_A, X_B are taken from the Pauling scale (e.g., $F = 4.0, O = 3.5, Cl = 3.0, Na = 0.9, Cs = 0.7$).

4. Dimensional Analysis in Mole/Density Calculations

- Always convert units upfront:
 - $1 \text{ nm} = 10^{-9} \text{ m} = 10^{-7} \text{ cm} = 10 \text{ \AA}$
 - $1 \text{ cm}^3 = 10^{-6} \text{ m}^3$
 - Atomic weight A in g/mol, Avogadro's number $N_A = 6.022 \times 10^{23} \text{ atoms/mol}$.